Release Notes

CADP internal QA build release notes — generated from OpenProject. Not the in-app Release Notes feature used by tenants.

1. Release Overview

Field	Details
Product Name	**CADP Workspace**
Release Version	**Release 1.0.0 (28)**
Release Date	**October 31, 2026**
Release Type	**Feature & Bug Fix Release**
Prepared By	**Development Team**
Approved By	**Ganagavathy**
Environment	**QA**

2. Release Summary

This release of CADP Workspace Version 1.0.0 (28) delivers 6 bug fixes, 10 new features, and 10 enhancements across the platform. Highlights include Language Mapping setting (where the workspace language applies), Platform entries, and Platform screen catalog. It also includes a required configuration change (see Configuration Changes). Refer to the Known Issues section for current limitations.

3. New Features

Feature ID	Feature Name	Description	Impact
	Language Mapping setting (where the workspace language applies)	New `languageMapping` field on the workspace document and; `GET`/`PUT /workspaces/:workspaceId/language-mapping`. It stores which layers show in the; workspace language (`applyTo.platform` / `workspace` / `data`) and, per feature, which fields and; actions do (`features[featureId].items[componentId] = { kind, label, enabled }`). The language; itself is still `WORKSPACE_LANGUAGE`; the GET returns it with the provider for display. A PUT; replaces only the features it sends (`null` removes one). Nothing saved means everything; translates, which is the behaviour before this setting existed.	
	Platform entries	The language mapping also stores `platform` (CADP's own menus/screens, keyed by menu; node, plus `common-actions`), in the same entry shape as `features`, and returns it on GET. PUT; accepts `platform` the same way as `features` (`null` removes an entry).	
	Platform screen catalog	The mapping also stores `platformScreens`: the fields found on CADP's own screens, as; admins open them. Same entry shape, plus `path` on the entry and `type` on each item (Heading,; Tab, Field, Column, Placeholder, Button). GET returns it, and PUT accepts it like; `features`/`platform`. A `platform` entry may now carry `path`, which limits its switches to; that screen.	
	Routes for Workspace Settings › Language Mapping	New menu node "Language Mapping" (`submenu_015_language_mapping`) under Workspace; Settings, with View/Edit on `/rbac/master/workspaces/:workspaceId/language-mapping`. The GET is; also in the Dashboard's always-on View set, because every screen reads the mapping. Until menuData; is reseeded and roles are granted the new node, `authorise.middleware.js` aliases the GET/PUT to; the workspace environment-variables View/Edit right (whoever can change WORKSPACE_LANGUAGE can say; where it applies).	

Feature ID	Feature Name	Description	Impact
	Author-set run order for a trigger's "run all connected paths" next steps	Workflow authors can now set the order in which a trigger's next steps run when Branching Mode is "Run all connected paths". Connections carry an optional `order` number; the parallel walker sorts the fan-out by it (numbered steps first, ascending; unnumbered steps after them in saved order). Invalid values (non-integer or negative) are reported by the workflow lint as `CONNECTION_INVALID_ORDER`.	The steps run one after another, and a failing step stops the ones after it, so order matters (e.g. "GRN Reject": an upsert into the trigger's own feature ran first and changed the source record). Before this, order was the connection array order — which the save route sorts by connection id, i.e. effectively random.
	Run Order list in the trigger's Branching Mode section	When a trigger's Branching Mode is "Run all connected paths", a "Run order" list of its next steps appears below the dropdown, with up/down arrows. Reordering numbers the trigger's outgoing connections 1..N (`connection.order`), saved with the workflow. Template workflows keep `order` when loaded.	Lets authors control the sequence of the trigger's next steps; the backend runs them one after another in this order.

Feature ID	Feature Name	Description	Impact
	Workspace Settings › Language Mapping	New Workspace Settings section. It shows the workspace language (read-only; set under; Security › Environment Variables) and lets an admin choose where it applies: the Platform /; Workspace / Data layers, and per feature which fields and actions (toggle on = workspace language,; off = default language). The runtime translator (`documentTranslator.ts`) now takes an `allowed`; filter built by `languageMappingScope.ts`:; On a feature screen, inside `<main>`, a table cell or option is Data. A field or action label; switched off for that feature stays default. Everything else is Workspace.; Module and feature names are Workspace wherever they appear.; Everything else is Platform.; `tx()` labels follow the Platform switch. A saved mapping applies immediately via the; `workspace-language-mapping-saved` event.	
	Platform tab lists CADP's menus, screens and common buttons	The Platform side of Language Mapping was only a note. It now shows the platform menu; tree, built from the signed-in user's navigation permissions.; Selecting a menu lists it and its sub-menus/tabs under "Menus & Screens", each with its own; toggle.; "Actions" lists the buttons shared by every platform screen (Save, Cancel, Create, Delete…).; Saved under `platform` in the mapping (one entry per menu node, plus `common-actions`).; At runtime a platform label switched off stays in the default language, both on the page; (`languageMappingScope.ts`) and in `tx()`.	

Feature ID	Feature Name	Description	Impact
	Platform screens list their fields	CADP's own screens have no field registry, so the app records their fields when a; screen is opened.; **Who records:** only a user who can edit Language Mapping (or, until RBAC is reseeded,; environment variables). On a platform screen, once the page has been quiet for 2s,; `platformScreenCapture.ts` reads it.; **What is read:** headings, tabs, labels, column headers, placeholders and buttons inside; `<main>`, in English even if the page is showing a translation. Table cells and; `data-no-translate` content are skipped.; **Where it goes:** into `platformScreens`. It is keyed by route pattern (ids → `:id`) plus the; section heading, and saved again only when new fields appear.; **Platform tab:** lists these under "Screens", grouped by first route segment, each with its; own Fields and Actions.; **Toggles:** saved under `platform` with the screen's `path`. They apply only on that screen; (`isPlatformLabelOff`). Menu and common-button switches still apply on every platform screen.	
	Platform menus list every field of their screen	Selecting a platform menu in Language Mapping now lists the fields its screen; shows, not only its tabs:; field labels, placeholders, column headers, headings, tab and option labels, and buttons; (under Actions); live items, one row each:; workspace environment variables and their groups and descriptions; workspace password policies; tenant variables and policies; platform variables and password policies	

4. Enhancements / Improvements

Enhancement ID	Module	Description	Benefit
	data-import	The XLSX sample template now gives a table component's own dropdown columns (manual; options) a selectable in-cell dropdown on the table's sheet, the same way manual dropdowns on the; Records sheet already worked. Option values go on a per-table `<Table> Options` sheet.	`getSampleTemplateStructure` only filled table-sheet `columnOptions` from; `buildReferenceTableColumnOptions` (a reference table's match columns), so every manual-dropdown; column inside a transaction/data table got `null` and no dropdown. New; `buildTableManualColumnOptions` / `_getManualTableColumnOptions` read the column's options from; root/basic/advanced/componentProps (mirroring the frontend's `resolveColumnOptions` /; `resolveColumnOptionSource`), only for a `manual` source — users/roles/org and feature sources are; skipped. Reference match-column options still take precedence for the same column.

Enhancement ID	Module	Description	Benefit
	formula-aggregate-filter-1762	A filtered aggregate — `COUNT_RECORDS({field}, {status} == "closed")` and the rest of the `*_OF_RECORDS` family — now applies its filter against the **records** rather than a flat list of the field's values, and an aggregate chosen from **"Saved records"** on the current feature now resolves its source instead of reading nothing. `== 0` no longer matches every record, `!= 0` no longer matches none, and `> N` no longer matches nothing.	Two defects combined, and both had to be fixed for a filtered count to be right.; 1. **The array path won.** The external resolver replaces an aggregate's field token with `__aggN`, a flat list of that field's values — fine for an unfiltered total, but a filter's tokens name *fields of a record*, and a list of values has no record to read them from. The filter tokens resolved to nothing and were substituted as `0`, so every comparison was really against zero.; 2. **No feature resolved.** `pickAggregateFeatureId` consulted only `selectModuleSets`, which lists the features a user explicitly added. A same-feature aggregate picked as "Saved records" emits no set at all — the token map is what records it, as `source: 'external'` plus the owning feature — so no feature resolved and the aggregate's records were never fetched.; Observed on DEV before the fix, over 4 records all with non-zero values: `== 0` → 4, `!= 0` → 0, `> 0` → 0.

Enhancement ID	Module	Description	Benefit
	formula-aggregate-view-scope	A formula aggregate (`COUNT_RECORDS`, `SUM_OF_RECORDS`, and the other `*_OF_RECORDS` functions) now also honours the role's **View** scope — `Own` or `Group` — not just its Data Access Control filter. Two paths are covered: the cross-feature record scan and the same-feature aggregate push-down pipeline. The aggregate and the user's own record list now report the same set.	Two different mechanisms narrow a feature's rows, and they live in different places. The DAC **rule** already reaches every read, including aggregate pipelines, through the global plugin (`config/db/session-param-plugin.js`). The `view` **scope** is not a rule — it never appears in `permissionIndex.filter` — so it is applied by the route as `_metadata.createdById`. A formula has no route to inherit it from, so a `View: Own` user's aggregate counted every creator's records. Measured on a test feature: a role set to `View: Own` was shown the full workspace total while its own list returned only its own rows.
	formula-time-diff-arg-type	`TIME_DIFF({field}, "10:00", "hours")` now fails validation when `{field}` is a **Text** field, with `TYPE_MISMATCH` on argument 1. A time written as a quoted literal — `TIME_DIFF({end}, "10:00", "hours")` — and a real TIME field both continue to validate as before.	The validator's time check exempted any `string` argument, because a quoted literal such as `"10:00"` is how a time argument is normally written. But a Text field also resolves to `string`, so `TIME_DIFF({city}, ...)` passed validation and then produced `NaN` at runtime — the error surfaced as a wrong number in the form rather than as a validation message. The two cases are distinguishable in the AST: a quoted time is a `ConstantNode`, a field reference is not. The exemption is therefore narrowed to literals rather than removed.
	ai-welcome-icon-a11y	The badge icon on the AI assistant's welcome card now shows a tooltip on hover and announces itself to a screen reader. The text names what the panel is for: **Create with AI**, **Ask about the platform**, or the read-only variant's own title.	The icon was rendered `aria-hidden` inside a bare `div`, so it had neither a hover tooltip nor an accessible name — QA filed it as "FAQ icon does not display tooltip on mouse hover" (#1740). Naming it is also the honest thing for a screen reader: the icon states the panel's purpose, and a decorative hidden icon states nothing.

Enhancement ID	Module	Description	Benefit
	ai-widget-thread-per-tab	Switching the AI assistant between the **Create** and **Ask** tabs now returns you to the conversation you were last in on that tab, instead of blanking the transcript back to the welcome card.	The tab switch set the active thread to `null` outright, so the transcript fell back to the empty welcome state until a manual refresh — even though the thread still existed (the server held it for Create, `localStorage` for Ask) and **Show chat history** still listed it (#1726). The thread was never lost; the widget just stopped pointing at it.
	formula-wizard-aggregate-source	An aggregate built through the guided **Formula Wizard** now reads the stored records of the feature it names — including when that feature is the one the formula lives on — and a value typed into the wizard's condition no longer silently matches nothing.	Two defects, both on the wizard path only (the field picker was already correct).; 1. **Wrong token source (#1441 / #1623-#1638).** The wizard stamped `source: 'local'` on every token belonging to the current feature. For an aggregate that is wrong: "This form" holds exactly one value, and aggregating one value is meaningless, so the engine had no record set and returned its empty fallback — a same-feature aggregate read `0.00` however many records existed. The field picker has always carried this distinction (`readsSavedRecords` → `source: 'external'`); the wizard did not, so the guided path could not produce a working same-feature aggregate at all.; 2. **Double-quoted values (#1762).** A value typed into the wizard was wrapped in quotes unconditionally, so typing `"Biscuits"` — quotes included — produced `""Biscuits""`, a literal that matches nothing. The condition was valid and simply never matched.

Enhancement ID	Module	Description	Benefit
	formula-year-integer-display	A formula whose entire expression is `YEAR(...)` now displays as a whole year — `2026`, not `2026.00` — in the form field, the table cell and the list/view. The rule is anchored to the whole expression: `YEAR({d}) + 1` and `ROUND(YEAR({d}) / 2)` are ordinary numbers again and keep their configured formatting.	The numeric display default is two decimal places, which is right for money and wrong for a calendar year (#1668). Three formatters disagreed about it — the shared engine formatter, the form field and the transaction table each had their own path — so the same formula could read `2026` in one place and `2026.00` in another.
	formula-year-special-case-revert	Removes the rule that made a bare `YEAR(...)` formula render without decimals regardless of the field's configuration. A year now follows the field's Number Format like every other numeric formula. The one part of the earlier change that is **kept** is the `decimalPlaces` clamp correction: a configured 0 now means zero decimals instead of silently becoming one.	The earlier change special-cased `YEAR(...)` so it always rendered as `2026`. That overrode an explicit design choice — the platform already lets a designer turn decimals off (Advanced → Number Format → uncheck Decimals), and the two-decimal default is applied to every numeric formula on purpose, which is why `3 + 2` reads `5.00`. Special-casing one function made a year behave differently from every other number without anyone asking for it, and it made the display depend on the expression's shape rather than on the field's settings. **#1668 is a configuration answer, not a defect.**
	image-upload-reject-message	When the image picker refuses a file before uploading it — too large, or the wrong format — it now shows a short message naming the file and the reason, e.g. `photo.png — File is larger than 10MB`. Previously the picker simply did nothing.	The size/format check happens before the file becomes an upload, so a rejected file never enters the preview list and had no thumbnail to hang an error badge on; the only trace was a `console.error` the user never sees (#1708). QA filed it as "6.68 MB PNG: not uploaded, no error/validation message" — the check was working, but silently.

5. Bug Fixes

Bug ID	Description
1162	Bullet List and Numbered List actions remove selected text instead of applying formatting.
1441	Formula field---> The formula field output is not displayed in the list data when the same feature variable is used. It only works when a different feature variable is mapped. This issue occurs only in the Aggregate field.
1571	Components → Calendar → Calendar Properties→ Calendar property settings are not reflected after Save and Publish.
1699	Components → Tooltip→ Tooltip is not working on the Create page.
1701	Components → Tooltip → Field Properties → Text→ Tooltip text field allows unlimited character input.
1865	Components→ Radio Button component name changes to Radio group.

6. Configuration Changes

Configuration	Description	Impact
OCR.Space key is read from the vault on deployed pods	`OCR_SPACE_API_KEY` is removed from `.env.production`, so deployed pods load it from the vault secret; `ocr-space-api-key` (mounted as `dev-ocr-space-api-key` or `ocr-space-api-key`) through the mapping that already; exists in `src/config/runtime-bootstrap.js`. `.env.development` / `.env.uat` keep the key for local runs (CSI off).	A value in the env file wins over the vault, so the vault mapping was never used and the live key sat in; git. **Deploy:** the `cadp-dev-kv` SecretProviderClass must expose `ocr-space-api-key` (or `cadp-dev-env` must set; `OCR_SPACE_API_KEY`) before this ships — otherwise OCR returns "OCR service is not configured (missing; OCR_SPACE_API_KEY)". The key has been in git history, so rotate it when it moves to the vault.

Configuration	Description	Impact
Workflow OCR node reads the server OCR.Space key from the vault	The OCR node's server-side OCR.Space key (`OCR_SPACE_API_KEY`) is now loaded from the vault at startup,; the same way client-ws loads it, so OCR nodes without their own API key use OCR.Space instead of falling back; to the lower-quality local Tesseract OCR. The service reads the mounted secret `dev-ocr-space-api-key`, then; `ocr-space-api-key`; a value already in the environment still wins. The key is optional — startup does not fail; without it.	The node falls back to `process.env.OCR_SPACE_API_KEY`, but this service had no vault mapping and no env; entry for it, so the server key was never set and every key-less OCR node silently used Tesseract; (`usedFallback: true`). Deploy: the `cadp-dev-kv` SecretProviderClass must expose `ocr-space-api-key` (the same; secret client-ws already maps) to the workflow pod.

7. API Changes

API Name	Type	Description
None	N/A	No API changes are included in this release.

8. Known Issues

Issue ID	Description	Workaround
	Role Management Change Effect	Because some menu structures have changed, certain features may return a 403 status code. To resolve this, edit the role, select the platform feature, and save the role again.
	Scheduler with Workflow Not Working	Use the conventional feature-wise approach wherever workflow scheduling is required.
	Formula Builder UI – Field Insertion Through Typing	This works in current fields. For external feature fields, it won't work, as there could be fields with same names from multiple features.
	Formula Builder – AI-Based Formula Writing/Editing Not Implemented	Use manual formula entry instead.

Issue ID	Description	Workaround
364	Organization Details → Basic Info: Website URL accepts invalid URL formats without validation.	Expected: Validate the URL format and show an error message for invalid values, such as www.example or example@123.www.example
	Builder Tool Not Available on Mobile Web	The builder tool is not supported on mobile web. This is implemented as expected — builder functionality is restricted to desktop browsers by design. It should be available on Tab though.
	Mobile – Formula Field Not Working in Tables & Charts	Formula fields are not evaluated correctly when used inside Tables and Charts on mobile. This is under review.

9. What to Test

- Login – Login, Sign In with Google (have to be configured in external providers in Tenants of metadata), Sign In with Microsoft (have to be configured in external providers in Tenants of metadata)
- **Create Application**
  1. Start from Scratch
  2. Choose/Create Vertical Template

- **AI-Assisted Creation & Ask FAQ** (Cavin's AI assistant — the AI launcher icon)
    1. **AI App Creation**
        a. From the workspace (any page other than Modules Management / Features Management / Builder Tool), open Cavin's AI and select the **Create** tab.
        b. Verify the helper line reads "Create builds a new application from your description".
        c. Describe the application to build and send. Verify creation progress is shown and "Application created from AI." appears on completion.
        d. Verify the generated application, its modules and features are listed and can be opened in the workspace.
        e. If the AI asks clarifying questions, answer them and verify creation continues (no success message is shown until something is actually created).
    2. **AI Module Creation**
        a. Open an application's **Modules Management** page, open Cavin's AI and select the **Create** tab.
        b. Verify the helper line reads "Create builds a new module here".
        c. Describe the module to build and send. Verify "Created N module(s) from AI." appears and the module list refreshes with the new module(s).
        d. Open the created module and verify it can be edited further.
    3. **AI Feature Creation**
        a. Open a module's **Features Management** page (or the Builder Tool), open Cavin's AI and select the **Create** tab.
        b. Verify the helper line reads "Create builds a new feature here".
        c. Describe the feature to build and send. Verify "Created N feature(s) from AI." appears and the feature list refreshes.
        d. From the Builder Tool, verify you are taken to the first created feature in the builder.
        e. Verify the generated form has appropriate components/fields, then publish it and verify records can be created, viewed and edited.
    4. **Create / Ask mode behaviour**
        a. In the Create tab, type a question instead of a build request. Verify nothing is created and an "Ask this in Ask mode" suggestion is offered; clicking it moves the question to the Ask tab.
        b. With a user who has only AI Builder or only AI Ask access, verify only that tab is available; with no create access, verify past conversations are read-only.
        c. Verify conversation history can be searched and reopened for each mode.

5. **AI Ask FAQ**

   Scope: AI Ask FAQ is trained only on the following platform documentation. Questions must be asked on these topics only:

   - CADP User Login Guide
   - CADP Functional Specifications
   - CADP Erudite Management Complete
   - CADP Workspace Guide
   - CADP User Preferences Guide
   - CADP Role Management
   - AI FAQ Guide Transactions Runtime
   - CADP Login Complete
   - CADP Erudite Management Complete Guide
   - CADP Features Guide
   - CADP Modules Guide
   - CADP Platform Architecture Report
   - CADP Templates Integrations Workflows Basics Guide
   - CADP Features And Submodules Guide
   - CADP Organization Guide
   - CADP Builder Tool Guide

   a. Open Cavin's AI, select the **Ask** tab and ask a question about platform configuration on one of the trained topics above.

   b. Verify a relevant and accurate answer is returned.

   c. Ask a follow-up or ambiguous question and verify it is handled gracefully.

   d. Ask a question outside the trained topics and verify AI Ask FAQ declines or returns an appropriate out-of-scope fallback rather than a fabricated answer.

   e. Verify Ask mode changes nothing in the workspace (it only answers).

- **User Management**

  1. Create/Manage Users
  2. Create/Manage Groups
  3. Import/Export Users
  4. Archive/Unarchive Users
  5. Locked Users
  6. Waiting Approval
  7. Please update existing users in each workspace.
  8. In the Users tab, when creating a new user, the Approval Type default is No Approval.

- **User Preferences**
    1. Profile update – first name, last name updating, profile picture updating.
    2. Change password
    3. Locale and Date-time updating
    4. Interface
        i. Theme change, color change. Color change here will override the brand color for that specific user of that workspace only.
        ii. Records per page is reflecting.
        iii. Language (currently only English accepted)
    5. Customization – module and features are visible based on visibility setting, quick access for that enabled feature is redirecting to feature transactions (records) page.
    6. Privacy & Security
    7. Miscellaneous – Default landing page, preferred currency, auto-save drafts, save interval
    8. Preferred user workspace
    9. Reset Preferences
- **Groups**
    1. Archive/Unarchive

- **Roles**
    1. Platform Roles
    2. Archive/Unarchive
    3. Workspace Role
    4. Feature Access Level
    5. Data Access Level
    6. User or Group Role Mapping
- **Workspace Management**
    1. Create Workspace
    2. Clone Workspace
    3. Archive Workspace
    4. Workspace Settings
- **Organization Management**
    1. Organization Details
    2. Organization Address
    3. Branding Details
- **Module/Feature Management**
    1. Create/Manage Module
    2. Create/Manage Features
    3. Design components for features
    4. Features and Modules can be published
    5. Audit Logs (Platform & Workspace)
    6. Version Logs (Workspace & Workflow)
    7. Staging
    8. Feature Card and Module Card Customization
- **Notification & Alerts**
    1. Email Notification & Templates – Create template and use template to send notification
    2. SMS Notification & Templates – Create template and use template to send notification
    3. WhatsApp Notification & Templates – Create template and use template to send notification
    4. In-App Notification & Templates – Create template and use template to send notification
    5. Push Notification & Templates – Create template and use template to send notification

- **Erudite Management**
  1. **Configuration (prerequisite)**
     a. Enable Erudite Management in Role Management before testing.
     b. Verify users without the role cannot access the Erudite Management module.
  2. **FAQs**
     a. Create a new FAQ with a question, answer, and category (Account, Billing, General, Technical).
     b. Verify the FAQ appears in the FAQ list after saving.
     c. Enable "Display this on the dashboard" and verify the FAQ is visible on the dashboard.
     d. Edit an existing FAQ and verify changes are saved correctly.
     e. Delete a FAQ and verify it is removed from the list.
     f. Search for a FAQ by keyword and verify correct results are returned.
  3. **Documentation**
     a. Upload a document (PDF or DOC, max 5 MB) with a name, description, and category.
     b. Verify the document appears in the Documentation list after saving.
     c. Attempt to upload a file exceeding 5 MB and verify a validation error is shown.
     d. Attempt to upload an unsupported file format and verify it is rejected.
     e. Enable "Display this on the dashboard" and verify the document is visible on the dashboard.
     f. Edit document details and verify changes are saved correctly.
     g. Delete a document and verify it is removed from the list.
  4. **Tutorial Videos**
     a. Upload a video (MP4, MOV, or MKV, max 50 MB) with a name, description, and category.
     b. Verify the video appears in the Tutorial Videos list after saving.
     c. Attempt to upload a file exceeding 50 MB and verify a validation error is shown.
     d. Attempt to upload an unsupported file format and verify it is rejected.
     e. Enable "Display this on the dashboard" and verify the video is visible on the dashboard.
     f. Edit video details and verify changes are saved correctly.
     g. Delete a video and verify it is removed from the list.
  5. **End User Access**
     a. Verify end users can browse and read FAQs, documentation, and tutorial videos.
     b. Verify items marked "Display on dashboard" appear on the user's dashboard.
     c. Verify items not marked for dashboard display do not appear on the dashboard.

- **Components**
    1. Layouts
        1. Column
        2. Section (various nested scenarios)
    2. Data Management (API Integration, Device Integration)
    3. Input Components (Phone Number, Password, Dropdown with Feature as Field Value)
    4. Specialized Input & Display
    5. Display and Interaction
    6. Media and Files
    7. Forms and Validation
    8. Utility Components
    9. Data Config Component
    10. Hyper Link Component
    11. Favorite Component
    12. OCR Component
        a. Add the OCR component to a feature form and upload an image or document.
        b. General text extraction → verify the text is extracted from the uploaded image/document and displayed within the selected output textarea
        c. Verify the extracted text is saved correctly with the record and is visible on View/Edit.
    13. Analytics & Report Components
        1. Chart Component
        2. Gantt Chart Component
        3. KPI Card Component
    14. Form UI
        1. Table component and Table level formulas, Summary row formulas
    15. Reusable Component
- **Page Builder**
    1. Create a new page and verify it is saved and listed.
    2. Drag-and-drop layouts (columns, sections) and reusable components onto the page canvas.
    3. Configure component properties and verify changes reflect on the page.
    4. Reorder, move, and delete components on the canvas.
    5. Edit an existing page and verify changes are persisted.
    6. Delete a page and verify it is removed from the list.
    7. Publish the page and verify it renders correctly at runtime.
    8. Verify the page is responsive across mobile, tablet, and desktop.
    9. Verify role-based access — only authorized users can build/edit pages.

- **Workflow Management**
    1. Visual canvas for creating workflows
    2. Drag-and-drop node placement
    3. Node connection with visual links
    4. Canvas zoom and pan controls
    5. Grid snap for alignment
    6. Create new workflows
    7. Edit existing workflows
    8. Delete workflows
    9. List/view all workflows
- **Workflow Nodes – Control Flow**
    1. Start Node (workflow trigger point)
    2. End Node (workflow termination)
    3. Condition/Decision Node (if-else logic)
- **Workflow Nodes – Communication**
    1. Email Notification Node
    2. SMS Notification Node
    3. User Task/Approval Node
    4. System Notification Node
- **Workflow Monitoring**
    1. Execution history logs
    2. Workflow instance status tracking
    3. Node-level execution details
    4. Error logs and stack traces
    5. Approval node changes for user selection
    6. Added SLA and escalation reminder for approvals
    7. Added parallel workflow approval node
- **Printing Templates**
    1. QR Code – Generate and Print
    2. Barcode – Generate and Print
    3. Invoice – Generate and Print
- **Role-Based Dashboards**
    1. Role-based dashboard template creation
    2. Dashboard rendering based on created template

- **Scheduler & Task Scheduler**
    1. Create and use Schedule
    2. Create and use Task Schedule
    3. Event-based task schedule
    4. Schedule-based task schedule
    5. Schedule a notification
    6. SMS
    7. Mail
    8. WhatsApp
    9. Push
    10. In-App
    11. Schedule a report send (PDF) through mail from a configured email template.
- **Formula Builder Functionality**
    1. Same Feature
        a. Arithmetic
        b. Logical
        c. Comparison
        d. Text
        e. Math
        f. Date
        g. Conditional
        h. Miscellaneous
        i. Aggregate → with filter conditions and without filter conditions.
    2. Cross Feature
        a. Arithmetic
        b. Logical
        c. Comparison
        d. Text
        e. Math
        f. Date
        g. Conditional
        h. Miscellaneous
        i. Aggregate → with filter conditions and without filter conditions.
    3. Incompatible datatype operations, for example, arithmetic on a text input field, should show a validation error.
    4. Unrecognized fields, functions, or operators should show a validation error.
    5. If external fields are called without where() conditions, it should show MULTIPLE RECORDS FOUND ERROR in preview for non-aggregate functions/operations.
    6. Variables are not working. Review which variables should be displayed in the modal.

- **Formula Builder UI**
  1. On clicking a feature from the feature list, fields in the selected feature should be displayed, except for the current field, to prevent self-call loops.
  2. On selecting external feature fields, where() should appear in the modal.
  3. On selecting where(), the modal should automatically switch to the Feature and Fields tab.
  4. On hovering over a field, field information such as module name and feature name should be displayed.
  5. The Clear button or backspace should work as follows:
     i. First click of clear/backspace will select the token.
     ii. Next click of clear/backspace will clear that token.
  6. Empty formulas should not be saved.
  7. Formulas should not be saved without validation.

- **Formula Builder – Helper Tools**
  1. **Cheat Sheet** (**?** icon)
     a. Open the Cheat Sheet and verify functions are grouped by category (Lookup, Nested Lookup, Aggregation, Conditional, Math, String, Date).
     b. Search for a function and verify the list filters correctly.
     c. Click the copy icon on an example and verify it is copied to the clipboard.
     d. Press Esc and verify the modal closes.
  2. **Examples**
     a. Open the Examples drawer and verify 50 scenarios across all categories are listed.
     b. Filter by a category and search; verify results update.
     c. Click "Use This Pattern" and verify the formula is inserted into the editor.
  3. **Test Formula (Playground)**
     a. For a current-feature calculation (e.g. *{number_field} * 5*), open Test Formula, verify a type-aware input appears per field (number/date/true-false/text), enter values, run, and verify the correct result, type, and (where relevant) matched-record count.
     b. For a lookup formula (uses *.where()* / external feature fields), verify NO value-input form is shown and a "Lookup formula – test against a real record" message appears instead.
     c. For an aggregate formula (SUM/COUNT/AVG/MIN/MAX of records), verify the "Aggregate formula – test against a real record" message is shown and the form is not available.
     d. Verify the Test Formula button is disabled when the formula is empty.
  4. **Template Library**
     a. Open the Template Library and verify the **Most Used** tab lists the 20 platform built-in templates (e.g. GST, TDS, net salary, lookups, aggregations).
     b. Search and filter by category; verify results update.
     c. Click "Use This Template" and verify the formula is inserted into the editor.
     d. With a formula in the editor, click "Save current formula as template", fill title/category/description, save, and verify it appears under **My Templates**.
     e. Delete a saved (non built-in) template and verify it is removed. Verify built-in templates cannot be deleted.
  5. **Formula Wizard**
     a. **Lookup mode:** choose "Lookup a value", select a source feature, select the field to return, add a match condition (verify the "Right side is a field reference" option shows a dropdown of the current feature's fields), check the live preview, and Insert.
     b. **Aggregate mode:** choose "Aggregate records", select feature/field, optionally add one filter condition, choose the function (SUM/COUNT/AVG/MIN/MAX), and Insert.
     c. Verify the formula preview only appears after a return field is chosen, and that going Back keeps your previous selections.
     d. After inserting, hover over the tokens in the editor and verify the correct Module name and Feature name are shown for each (including the source feature for the looked-up field).

6. **Smart Validation Messages**
    a. Enter an invalid formula (e.g. unknown function, missing where(), wrong argument count) and click Validate Expression; verify the error includes a helpful suggestion and an example, not just a generic message.

- **Integrations**
    1. Step 1: Go to the Integration Type field and select "Third-Party Software".

       Expected Result: A new dropdown field called Select Type should appear.
    2. Step 2: Check the options inside the Select Type dropdown.

       Expected Result: It should contain options including "Third-Party API" and "Google Drive".
    3. Step 3: Choose "Third-Party API" from the dropdown.

       Expected Result: The configuration fields specific to the Third-Party API should be displayed.
    4. Step 4: Change the selection to "Google Drive".

       Expected Result: The system should dynamically display the Google Drive configuration fields: Client ID, Client Secret, and Refresh Token.

10. What to Test on Mobile

- Login – Normal Login
- Login – SSO (Sign In with Google, Sign In with Microsoft)
- Workspace Switch
- Module List
- Feature List
- Components
- Feature Data – Create, View, Edit, Clone, Delete
- Dashboard
- Integration – Weighing Machine, Third-Party API (Budgie, SAP), and Payment Gateways (Stripe, Razorpay, and PayU)
- Theme
- Push Notification
- Workflow – Approve and Reject
- Charts – Feature Data
- Data Table – Feature Data
- Security – Prevent copy, paste, cut, and text selection